



examGenome CheatSheet on Algebra

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1	What is a ratio? Answer: A ratio compares two or more quantities by showing how many of one thing there are for a certain number of another. Explanation: It's written as 3:2 or '3 to 2'. Ratios are used to compare things like red marbles to blue marbles, or boys to girls in a class.
2	What is a proportion? Answer: A proportion is when two ratios are equal, like $1:2 = 2:4$. Explanation: Proportions show that two comparisons are the same. They are often used to find missing values in recipes, maps, or prices.
3	What is a rate? Answer: A rate compares two different kinds of quantities, usually involving time—like speed (km per hour) or cost (dollars per kilogram). Explanation: Rates always include 'per'—for example, '60 kilometers per hour' tells you how fast something moves.
4	What is a unit rate? Answer: A unit rate is a rate where the second quantity is 1—like '5 km per 1 hour' or '\$2 per 1 apple'. Explanation: Unit rates make it easy to compare prices or speeds. For example, which is cheaper: \$4 for 2 kg or \$3 for 1 kg? The unit rate shows \$2/kg vs. \$3/kg.
5	Do ratios, proportions, and rates have units? Answer: Ratios have no units (they compare the same kind of thing). Rates have units (they compare different things, like km/h). Explanation: In a ratio like 4 apples : 2 oranges, units cancel. In a rate like 60 km/h, the units (km and hour) stay because they're different.
6	What is the difference between direct and inverse proportion? Answer: In direct proportion, both quantities increase or decrease together. In inverse proportion, when one goes up, the other goes down. Explanation: Direct: more hours worked = more money earned. Inverse: more people working = less time to finish a job.
7	How do you check if two ratios form a proportion? Answer: Use cross-multiplication: if $a/b = c/d$, then $a \times d = b \times c$. Explanation: For example, $2/3 = 4/6$ because $2 \times 6 = 12$ and $3 \times 4 = 12$. If the cross products are equal, it's a true proportion.
8	What is a scale factor in proportion? Answer: The number you multiply by to get from one ratio to another equivalent ratio. Explanation: If 1:3 becomes 4:12, the scale factor is 4—because $1 \times 4 = 4$ and $3 \times 4 = 12$.

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9	Can proportions be used with decimals or fractions? Answer: Yes! Proportions work with any numbers—as long as the two ratios are equal. Explanation: Examples: $0.5/1 = 1/2$ or $\frac{3}{4} = 6/8$. The type of number doesn't matter—equality does.
10	Why are rates of change important in real life? Answer: They help us understand how things change over time—like speed, growth, or how fast a tank fills with water. Explanation: Rates of change turn real-world situations into numbers we can compare and predict—like 'This car is faster because it goes 80 km/h, not 60 km/h.'

